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College Exam Grader using LLM AI models

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***Abstract*— By far, the most effective knowledge assessment in college education is to give students exam and grade their answers then assess their level of understanding. However, exam grading can be time-consuming, tedious, cumbersome, and sometimes the grading results are not consistent with the rubric. Here, we propose an AI based exam grader that can not only ease educators' burden but also produce accurate, consistent, and precise grading results. We have used GPT-3.5, GPT-4.0, and Gemini-pro, respectively, as our grading engine. To verify the correctness, precision, and accuracy of our proposed grader, the results were compared with the instructor's grading result and also with human grader such as teaching assistants. In our experiment, GPT-4.0 showed the most reliable and consistent results.**

***Keywords*—ChatGPT, Gemini, grading, college education, artificial intelligence, prompt engineering**

I. Introduction

Assessment (also, "evaluation" is used in the same meaning) refers to measuring students' understanding of learning using various means. This is one of the most important elements of education, and through assessment, instructors can not only measure the level of understanding of individual students, but also measure the effectiveness of their teaching by checking the overall level of understanding of the teaching group. Based on such assessment, an instructor can not only improve student's learning way, but also the instructor's teaching method. Therefore, assessment is viewed as one of the most important elements in education as it is helpful for both instructors and students [1]. At school, assessment has been done in the form of exam, homework or quiz. These are the same in nature, with the student writing answers to given questions. Categorizing or quantifying students' answers in the assessment is called grading or scoring. (Here, the term "grading" will be solely used.) In most of colleges in the world, the grading procedure is mainly done by human, typically instructors or their teaching assistant. However, reading and evaluating students' answers is time consuming and leads to overwork for instructors. According to a 2014 report, professors in the U.S. colleges worked 61 hours a week, and 11% of that time was spent on course management, including grading and administrating course webpage [3]. In other words, professors in the United States spend, on average, nearly 6 hours a week on grading. According to this report, only 3% and 2% of the time was spent on primary research and writing, respectively. Based on this report, American professors seem spending more time on grading than on research, which is viewed as their primary role. To help easy their workload, many universities assign graduate students as TAs to courses for grading, but in many cases, professors themselves are responsible for grading directly due to financial issues or a lack of graduate students. In addition to the time spent on grading, to maintain fairness, objectivity, and consistency by excluding the grader's subjectivity in grading is another challenge [3]. Grading is always prone to student's complaint.

From the 2010s, the full-fledged AI technology has appeared and there comes out a glimmer of introducing such technology in educational fields. In particular, Large Language Models (LLM), including ChatGPT, have shown the ability to surpass human in understanding human questions and providing proper answers . Education field is also experiencing drastic changes. AI, especially ChatGPT has brought both negative and positive impacts in education. Until now, the negative side becomes apparent: students began to use ChatGPT in doing their homework and the goal of homework or take home exam becomes meaningless. However, the positive side of using AI in education is still under speculation. But we have noticed that AIs, especially LLMs have an ability to understand a text and compare this with other texts. If we can assure this ability comes with fairness, objectiveness, and consistency, college instructors can take advantage of this ability and save much time in grading students' answers. Considering this, we created an AI grader using APIs of ChatGPT and Gemini, two of the currently most used LLMs. With this, we have conducted various experiments to explore whether AI grading can be applied in practice. In this paper, the term "LLM" and "AI" are used in the same meaning. The outline of our paper is as follows: Sect 2. presents literature review, Sect 3. details our app's components and how to use, Sect 4. describes the experiments using our app and the result data, Sect 5. presents our conclusion and finally Sect 6. discusses the several issues in AI grading.

II. Literature Review

A. Introduction of AI into Education

From the late 2010s, there have come many studies to discuss AI in educational. Holmes et al. reviewed then-existing AI technology in education and explored possibility of introducing AI to various fields of education in both instructor's side and student's side. He predicted that AI, virtual reality, and augmented reality will be introduced into the educational field and change the aspect of education drastically in the near future. He also pointed out the problems that may arise with the introduction of AI in education field. He expected that ethical issues will arise when AI evaluates students instead of human instructors. Also colonialism issues will occur because AI education systems developed in developed countries can be used to implement in developing countries' education system, which may inject values of developed countries into students in developing countries. Finally, he expected commercialization of education will be expedited due to AI [4].

Yu-Ju Lan et al explored the future role of Generative AI (GAI) based on LLM in education. She pointed out the case that some students abused LLM for homework or solved problems without their own efforts. However, she also suggested that teachers actively make their teaching plan by the assistant of ChatGPT or provide students with personalized learning, eventually leading to creativity. She also predicted that educational AI would become the instructor's clone which can help students constantly beyond temporal and spatial limitation [5].

With the advent of ChatGPT, there have been many happenings in the educational field. In particular, students abusing ChatGPT for homework or take home exams is common all over the world, not only in English-speaking regions but also in native language speaking regions such as Korea and Hong Kong [6] [7] [8]. In this respect, it is clear that ChatGPT affects negatively education fields. To prevent such a cheating in online exam or take home exam, Ryznar proposed devices such as video proctoring, exam software, time-limits, and introduction of time-limiting and supervisory software [9].

B. *The Ability of LLMs to Understand Text*

Since the advent of ChatGPT in November 2022, there have been several articles in the media or academic studies about the ability of ChatGPT which made fair achievement in academic test or professional exams such as bar or medical doctor qualifications. Although ChatGPT's performance varies depending on the nature of the test, it was reported that there occur some hallucinations in answering to the given question, which refers to the phenomenon of saying something that not exist as if it exited [5].

Jang et al. studied LLM's consistency in conducting experiments which evaluated semantic, negation, symmetric, and transitive consistency in GPT-4 and ChatGPT (OpenAI's chatting web site). He concluded that ChatGPT shows a considerable level of language understanding and deductive reasoning, but its performance in various consistency does not reach the expectations [11].

C. *Grading and AI*

In 2017, a commercial grading solution using AI, Gradscope ®, was released by a team of instructors. The company claims that handwritten exam or homework papers can be graded using artificial intelligence. They said that their solution assures speed, consistency, and flexibility in grading. Here, consistency means that grading is fair and not varied over time, flexibility means instructors can have the control over the rubric of the exam or homework. They showed that students' handwritten code was successfully graded by their solution [4]. However, according to the video clip posted by the company on YouTube, the grading using this solution fell short of the grading using LLM. They seemed to focus more on recognizing students' handwriting or drawings and digitizing them [10].

Holmes reviewed many AI technologies in educational fields and classified them into 3 categories depending on their current feasibility level: "speculative", "researched" or "commercially available". He classified "Automatic Summative Assessment" (assessment process in instructor's side) as "researched" or "commercially available" , but pointed out that its introduction to fields remains controversial in assessment in high-stakes. Also, in his research, "Automatic Formative Assessment" (assessing a student's work and giving feedback) was classified into the same category , but pointed out that there is few commercial case and has difficulty in giving actionable feedback. Considering his evaluation on AI-based assessment, it seems that although there were some models then-available for AI in grading, the overall reliabilities of them were limited. Of course, it should be noted that his study was done before the advent of ChatGPT [4].

So far, the use of ChatGPT in education generally has raised issues on students' misuse or on its ability to make a proper answer to a specific question, but the use of ChatGPT as teacher's assistant has not been reflected on or studied much. Ellis tried a use case of ChatGPT for grading students' essay in the chatting mode of ChatGPT. He introduced a method of grading essays. He input a prompt to ChatGPT who supposed to be an expert educator, and let it read a rubric containing the grading criteria, then gave ChatGPT the student's essay and order to grade this essay [1].

III. FEATURES OF THE PROPOSED LLM GRADER

A. *The Component of Our Grading APP*

Our proposed system is illustrated in the Figure 1. The system utilizes the currently most popular LLMs: Open AI's GPT 3.5/4.0 and Google's Gemini-Pro. It is a web-based application that uses APIs of LLMs. A instructor logs in the system and input necessary things: the exam questions, accompanying rubrics and students' answers. Our system grades answer sets in an exam at once per student. Therefore,

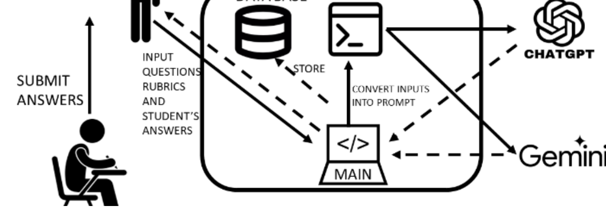


Figure 1. system flow of the proposed system

each time our system grades, a student's final grade (or score) is returned immediately. Figure 1 shows the diagram of the app.

After the instructor's input, the app constructs a prompt for each questions and invoke all three models for grading. Each LLM grades a student's answers according to the given rubric. The grading results are sent back to the instructor for review. (Figure 2)

The result of grading Joe Smith's Exam

No.	LLM	Content
1	Score (GPT35)	3
	Comment (GPT35)	Score:3 Reason: Two of the three largest countries, Russia and Canada, are in the answer.
	Score (GPT40)	2
	Comment (GPT40)	Score: 2 Reason: The answer includes two of the three largest countries in the world, but not in the correct order.
	Score (Gemini)	2
	Comment (Gemini)	Score: 2 Reason: Only two of the three largest countries are in the answer, so it deserves 2 points.

Figure 2. Screenshot of the output of the AI grader APP. The given question was "List 3 the 3 largest countries in order". The input answer was " Russia, China, and Canada"

B. *The structure of main prompt*

```
The given question is {Q}
and evaluation criteria is
{R}. The student's answer
is {A}. Evaluate based on
the given criteria in
"Score: integer" and give
the reason of the
evaluation briefly.
```

Figure 3. our prompt used in the AI grading app. {Q}, {R}, and {A} are question, rubric, and answers.

In the AI context, prompt refers to information, sentences, or questions which a user inputs into an AI for a specific purpose [1]. An LLM is a kind of AI where a user can use natural language as prompts. But natural language can have logical ambiguity in nature. This point can affect the grading quality of our app. Therefore, the prompt used in this app should be carefully designed to exclude the logical ambiguity. The structure of the prompt is shown in Figure .

C. *Importance of Rubric*

A rubric refers to a criterion for evaluation on an answer. In college education, the evaluation of homework or exam is made by the instructor himself/herself or TA. In most cases, the rubric, or grading criteria for problem set, is not given clearly in written form. Moreover, TA's grading is also largely left to the TA's discretion. However, different from a human TA, LLMs operate according to only given prompts. As a result, it is important how to write a rubric as a prompt clearly and without logical contradictions in AI grading. Much consideration must be given to writing a rubric.

A rubric consists of several conditional statements. For example, if a question says "list the 4 smallest prime numbers", then a rubric to this question may say "If the answer lists 2,3,5,7, then give 10 points. If it lists only three of them without any other number, give 5 points. Otherwise, give 0 point." There are 3 conditional ("if" or "otherwise") statements and these 3 cases covers all possible answers.

There are several cases in how a rubric covers the possible answers (Figure). Suppose we have a question and its rubric and a grader can give 0,5, and 10 points to a student's answer according to the rubric. Let Ω be the entire set of possible answers to a given problem.

$$\Omega = \{a_1, a_2, \dots, a_n \dots .\}$$

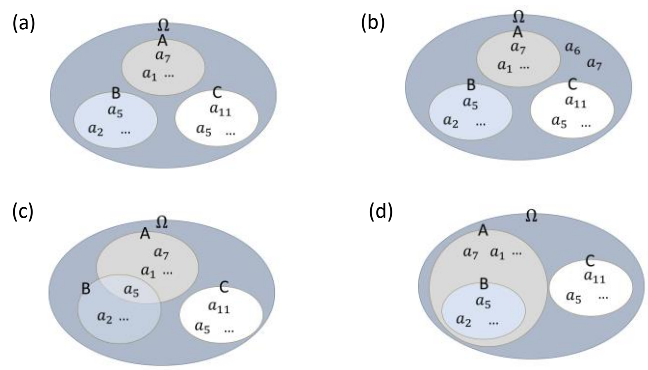


Figure 4. illustrations for how a rubric covers answers

Let a_i's are possible answers to the given question. (the number of a_i's may be infinite, since a student would not write exactly the same sentence as another student.) We can view a_i's as logical propositions. And let A,B and C be Ω 's subsets which corresponds to the point 0, 5 and 10. For example, an answer a_{11} is included in set C, then the student who submitted a_{11} can get 5 points in this problem. We should keep in mind that A, B, and C can be expressed in a conditional statement in the rubric. They can be viewed as truth sets for propositions a_i's. The good and bad conditional statements in the rubric are described in Figure 4. In (a), A,B, and C cover all the possible answers and there is no intersection among them. But in (b), $a_6, a_7 \notin A \cup B \cup C$ and this means that answers a_6 and a_7 are not graded properly. If we take an example with the given question, a rubric in case (b) says " If the answer lists 2,3,5,7, then give 10 points. If it lists only three of them without any other number, give 5 points. " and no "otherwise" statement. In this rubric, the answer with {6,9} cannot be graded properly. In (c), the intersection $a_5 \in A \cap B$ and the LLM grades a_5 according to A or B arbitrarily. In (d), $B \subset A$ and the grade can be inconsistent. If we take an example with the given question, a rubric in case (d) may say "If the answer lists 2,3,5,7 then give 10 points. If it includes three of them then give 5 points." Even if the answer is {2,3,5,7}, an LLM may think it can be also true in the second conditional "if" statement whose is satisfied with three numbers of them. Then the LLM may give the right answer only partial 5 points . As shown in this picture, the conditional statements in the rubric must be logically clear and cover all possible answers.

In set theory terms, the truth sets which are represented by the conditional statements in a rubric satisfies the below conditions. Let G_i $(i = 1,2, \dots n)$ be the truth sets for conditional statements in a rubric. Then,

$$G_1 \cup G_2 \cup \dots \cup G_n = \Omega$$

$$G_i \cap G_j = \emptyset \ \ if \ i \neq j$$

Even if the prompt to an LLM is written in natural language, the above mathematical condition should be always considered in writing a rubric. Therefore, the conditional statement of the rubric must satisfy all of the formulas of set theory above.

D. *One-shot Learning*

As an LLM has already studied numerous documents, it has a certain level of knowledge in any field. That's why ChatGPT can answer any question like an expert. However, college exams are sometimes graded based limitedly on the content taught by the instructor. No matter how correct the student's answer is, if it deviates from the course content, it is difficult to get full points. In this situation, we define evaluation based on given content as “in-class grading”, and evaluation based on a universal story as “out-class grading”.

In-class grading is an example of one-shot learning in LLM. One-shot learning refers a type of LLM learning that a user gives the LLM only one sample in a prompt. We tried grading with the following problems and rubrics to test whether our AI grading app performs in-class grading properly.

Question: Who are the three greatest novelists in the world?
Rubric: The most important novelists are Leo Tolstoi, Lu Xun, and Herman Melville. If the student's answer list 3 of them, give 5 points. If the student's answer list 2 of them, give 3 points. If student's answer list 1 of them give 1 point. Otherwise 0 point.

Figure 5 an example of question and rubric with “in-class grading”

Figure 5 shows an example of in-class grading. The question “who are the 3 greatest novelists in the world” is very subjective and can get many different answers. But, here, the instructor’s rubric teaches the LLM who they are and give the criteria to grade. Then the LLM will grade answers only based on this rubric, not on other information. In this way, we can grade answers in any subjective or instructor-specific matters, simply by detailing the instructor’s intention and emphasis. And after confirming that LLMs graded this question properly, we conducted the experiments.

IV. EXPERIMENTS

A. *Experiment Overview*

Several types of question are used in college, including multiple choice, matching, T/F choice, short answer, and essay question. For AI-grading, we chose short sentence answer for the experiment. In the problem which requires short sentence answer, students can have more flexibility in constructing answers compared to multiple choice and it does not give students unnecessary hints. In the problem which requires essay question, students should compose their answer composed of several sentences and their intelligent or cognition, including analysis, synthesis and understanding can be evaluated [2].

(**Question A**) What is the difference between guided and unguided transmission?
(**Rubric A**) Guided transmission refers to the transmission that data signals travels though physical media such as copper wiring or optical fibers that provide a specific path. Unguided transmission refers to the transmission data signals are carried in all directions through free space.
If the student’s answer is consistent with this sentence, give 5 points. If it is partially consistent, give 3 points. Otherwise give 0 point.
(**Answer A**) `The guided transmission is the transmission guided by an external object, and unguided transmission is the transmission which is unguided.`

Figure 2. Problem A, the rubric, and the mock-up answer

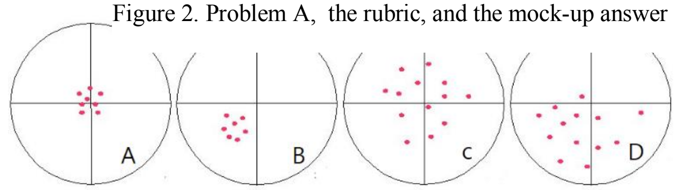


Figure 6. Accuracy and Precision (Source: Wikipedia)

These two types of problems are mainly used in tests at senior level and above and take more time than any other type for grading.

To evaluate the AI grading, we introduce the definition of accuracy and precision. In scientific experiment, accuracy refers to the closeness of a measured value to a standard or known value, and precision refers to the closeness of two or more measurements to each other [12]. In other words, grading with accuracy is correct and right, on the other hand, grading with precision is fair and consistent. To measure precision, variance is used, and to measure accuracy, percentage mean absolute error (MLE%) can be used [13]. Here, the smaller the variance is, the better precision the grader has and the smaller the percentage MLE is, the better accuracy the grader has. The accuracy and precision are illustrated case by case in Figure 6. A shows good precision and good accuracy. B shows good precision, but band accuracy. C shows bad precision, but good accuracy. D shows bad precision and bad accuracy. Good grading has both accuracy and precision. To evaluate LLM’s grading in precision and accuracy, we conducted Experiment 1 and 2.

B. Experiment 1: AI grading of mock-up answers

One of the biggest issues with LLM is that its generated answers to the same question can be varied on different attempt time. It means that a score generated by an LLM grading may be arbitrary and random. This means it may lack enough accuracy and precision. It is due to the nature of LLM which is based on countless random variables. [12].

In the first experiment, we chose 4 questions (A,B,C,D) and their rubrics from homework and exams in Towson University's undergraduate network class. A rubric includes a model answer (which can get full 5 points) for in-class grading. And it gives direction on how many points to give. (Figure 7) Each question is allocated 5 as full points. And each mock-up answer was written so that it should receive 3 partial points out of a full 5 points. Then with the four problems and the same mock-up solutions, we repeated AI grading 20 times.

The result is shown in Table 1. It shows mean scores and its variance for each problem in AI grading.

	GPT-3.5		GPT-4.0		GEMINI-Pro	
	Mean	Var	mean	var	mean	Var
A	2.85	2.02	3	0	3.10	0.20
B	4.70	0.53	3	0	4.10	1.04
C	3.30	0.53	3	0	2.80	0.80
D	2.40	0.88	1.95	0.05	2.30	0.96
Avg	**3.3**	**0.99**	**2.7**	**0.01**	**3.0**	**0.75**

Table 1 Mean and Variance of Scores

Table 1 shows that each LLM's mean score and its variance for each problem in 20 grading trials. Its last row shows the average values over the 4 problems. Each model's mean score to each question were generally the value (score 3) intended by the mockup answer designer, but GPT-3.5 and Gemini generate different score each trial. Only GPT-4.0 gave almost the same score 3 for all 20 trials. Out of 20 trials, only once score 2 other than score 3 was graded. From this experiment, we can say that only GPT-4.0 has guaranteed precision for grading.

	GPT-3.5	GPT-4.0	GEMINI-Pro
A	25%	0%	3%
B	57%	0%	37%
C	10%	0%	13%
D	20%	35%	23%
Mean	**28.0%**	**8.7%**	**19.2%**

Table 2. mean MLE% over 20 trials

As mentioned above, to measure accuracy, percentage MLE (MLE%) is sued. Table 2 shows MLE% of each MLL for each problem. Percentage MLE (MLE%) is defined as,

$$MLE\% = \frac{\sum|f_t - d_t|}{\sum d_t} \times 100$$

where, f_t is an observed value (here, it is LLM's generated score) and d_t is a true value (here, it is score 3, which is designed by the mock-up answer designer.). If MLE% is close to 0%, it means that each observed value is close to the right value. In our experiment, MLE% also varied depending on the problem and grading MML, but GPT-4.0 showed the best results. For problem D, GPT-4.0 had lower accuracy (35%) than other LLMs. (20% in GPT-3.5, 23% in Gemini-Pro) However, while other LLMs kept generating different values every 20 trials, GPT-4.0 consistently gave 2 points in 19 out of 20 trials. And the average MLE% over all problems is also the lowest in GPT-4.0 (8.7%). Therefore, its precision was much better than any other LLM.

C. Experiment 2: Human grading of mock-up answers

To validate the effectiveness of AI grading, we need to compare it to human grading in accuracy and precision. We assume that the precision in human grading is guaranteed. It is because human graders keep their memory while grading; the previous gradings always affect the next grading and he/she tries to keep consistency (precision). On the other hand, our AI grader app uses the chat completion method in their APIs. Chat completion is a python method that returns the LLM's answer to the given context and a question. Every time the computer instantiates this API module, the memory allocated to this method is renewed. It means the results of the previous grading does not affect the next grading, and as a result, precision in AI grading cannot be guaranteed. However, as shown in Experiment 1, GPT-4.0 nevertheless maintained precision.

On the other hand, the accuracy of human grading may not be guaranteed. This is because the degrees of literacy, understanding, and judgement vary among human graders. Considering these factors, we designed this experiment. We gave the same 4 problems and rubrics as in the experiment 1 to 7 graduate students who major computer science in Towson University and had them grade the same mock-up solutions as the app graded in the experiment 1. Then we calculated the percentage MLE (MLE%) of their gradings on all 4 problems and compare this to AI grading's MLE%.

	P1	P2	P3	P4	P5	P6	P7	**Avg**
MLE%	58%	41%	25%	17%	42%	33%	42%	**36.9%**

Table 3. Percentage MLE of Human grading

In Table 3, P1, P2, .., P7 refer to the 7 participants. Each cell shows a participant's averaged MLE% over 4 problems and the far right cell shows their average value. We can understand the accuracy of human grading varies from person to person, and comparing their average value to the values in the last row of Table 2, we can think that AIs' grading accuracy is no worse than human TAs accuracy.

D. Experiment 3: Recognizing Equivalent Meaning with Changed Expression

We designed the third experiment to check whether the LLMs used in our app would properly understand the meaning of sentences despite the changed expression. In this experiment, participants were told to read the mock-up solutions and rephrase them as much as possible without changing its meaning. Then, the changed answers were input into our AI grading app and graded five times to generate statistics. 4 graduate students who major computer science participated in this experiment. Figure 8 shows 4 rephrased mock-up answers to problem A.

(original mock-up answer) The guided transmission is the transmission guided by an external object, and unguided tranmission is the transmission which is unguided.

(Rephrased A-1) Guided transmission means a transmission that is controlled by an external object. unguided transmission is not controlled by an external object.

(Rephrased A-2) Guided transmission means the data is travelling through physical media while unguided transmission means data signals travels in all directions.

(Rephrased A-3) When an external object guides a transmission, it is called a guided transmission otherwise called as unguided transmission.

(Rephrased A-4) the guided transmission is guided by an external object, whereas unguided transmission is not guide by any external object.

Figure 3. Rephrased mock-up answers to Problem A

	R1			R2			R3			R4		
	G35	G4	GE	G35	G4	GE	G35	G4	GE	G35	G4	GE
A	2.4	3	3.4	3.4	3	1.2	1.8	3	2.8	3.0	3	4.2
B	5.0	3	3.8	5.0	3	4.6	5.0	3	4.2	5.0	3	3.6
C	3.0	3	3.0	3.8	3	2.2	3.0	3	4.6	3.8	3	3.4
D	2.2	2	2.6	2.2	2.2	1.4	2.6	2	2.6	2.7	2	1.8
AVG	**3.2**	**2.8**	**3.2**	**3.6**	**2.8**	**2.4**	**3.1**	**2.8**	**3.6**	**3.7**	**2.8**	**3.3**

Table 4. the scores of AI gradings on rephrased answers

Table 4 shows the results of the third experiment. Here, R1, R2, R3, and R4 refer to 4 participants. G35 refers to GPT-3.5, G4 refers to GPT-4, and GE refers to Gemini-pro, and each row represents each problem A, B, C and D. A score in a cell is the average score of 5 grading trials by each LLM's grading. As mentioned above, the original mock-up answer is designed to get 3 points in the given rubric. Therefore, if the rephrased meaning is not changed, it should get 3 points, too.

Comparing the results in Table 5 with the results in Table 1, we can see that GPT-4.0 evaluates the meaning the same as the original sentence even when rephrased sentences were given. Moreover, GPT-3.5 and Gemini-pro also showed similar values to the results in Table 1. Each LLM seemed to understand the rephrased sentences similarly like the original one understanding. This shows that the grading by AI is consistent over many answers with the same meaning even if the detailed expressions in the answers are varied.

	R1		R2		R3		R4		Average	
	var	mle	var	mle	var	mle	var	mle	var	mle
GPT3.5	0.75	28%	0.8	33%	0.875	30%	0.6	30%	0.76	30.2%
GPT4.0	0	8%	0.05	6.6%	0	8%	0	3%	0.01	6.4%
Gemini	0.7	13%	1.375	48%	1.5	35%	2.0	38%	1.39	33.5%

Table 5. MLE% of AI grading on rephrased answers

The value in the last column of Table 5 is the average of the variance and MLE% of the four participants' answers graded by each LLM. We already mentioned variance is used as a measurement for precision, and percentage MLE is used as a measurement for accuracy. In this experiment, GPT-4.0 showed the best accuracy and precision among LLMs, and generated the same scores in the rephrased answers as in grading the original answers. Considering this, we concluded that GPT-4.0 is trustworthy enough to be used in grading an actual exam.

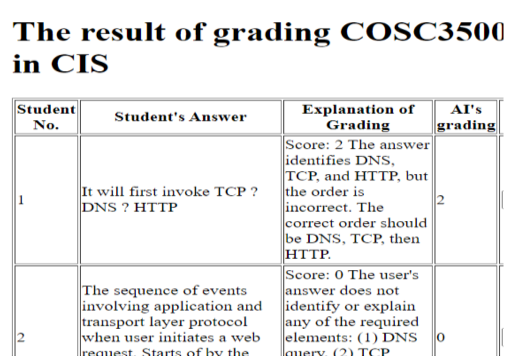
E. Actual Grading

Through the above experiments, we concluded that GPT-4.0's grading is trustworthy and put it into practice. For this purpose, our app was modified so that it use only the GPT-4 API and generate the grading points with the reason for such grading. In March 2024, there was the midterm exam in the undergraduate network theory class in Towson University. The exam was paper-based and 17 questions given to 59 students. Among them, one question is to ask the students to explain in sentences. We applied our app to grade students' answers of this question. Students could receive four levels of scores from 0 to 3 based on the rubric. (Figure 9)

(Question) Describe, step by step, the sequence of events involving application and transport layer protocols, when a user initiates a web request by clicking on 'www.towson.edu' in a web browser, assuming no prior cached data for web or DNS objects. Protocol involved: HTTP, DNS, TCP
(Rubric) IF the answer identifies (1) DNS query, (2) TCP channel establishment, and (3) HTTP request in this order and explains correctly all of them then give 3 pts. Or if the answer identifies (1), (2), and (3) but their order is wrong, give 2 pts. Or if the answer identifies (1), (2), and (3) but one of their descriptions are wrong, then give 2 pts. Or if the answer identifies (1), (2), and (3) but two of their descriptions are wrong, then give 1 pts. Or, If the answer identifies none of (1), (2), and (3), then give 0 pts. Or, if the answer is blank or nothing, give 0 pts.

Figure 4. The question for AI grading

Since the exam was paper based, the TA typed each student's answer into a document file for input. In order to ensure fair grading, student's typography or grammatical


The result of grading COSC3500 in CIS

Student No.	Student's Answer	Explanation of Grading	AI's grading
1	It will first invoke TCP ? DNS ? HTTP	Score: 2 The answer identifies DNS, TCP, and HTTP, but the order is incorrect. The correct order should be DNS, TCP, then HTTP.	2
2	The sequence of events involving application and transport layer protocol when user initiates a web request. Starts of by the	Score: 0 The user's answer does not identify or explain any of the required elements: (1) DNS query, (2) TCP	0

Figure 5. Screenshot of the AI grader output

mistake was entered intact without correcting them. The screenshot of AI grading are shown in Figure 10.

In addition to AI-grading, the TA also graded the students' answers and compare this with AI grading. The results is of a confusion matrix form in Table 6.

		Human TA			
		0	**1**	**2**	**3**
GPT -4.0	**0**	12			
	1		5		
	2		2	20	
	3		2	1	17

Table 6. Confusion Matrix in Experiment 3

In this table, the column number represents the score which the TA graded, and the row number represents the score which the GPT-4.0 graded. For example, the cell with the row number 2 and the column number 1 has number 2. It means that the number of students who get 2 in the human grading and 1 in the AI-grading is 2. The diagonal cell entries (12,5,20, and 17) are the number that the TA's score and the AI's score match. The coincidence rate is 54/59=91.5%.

There are 5 answers where the TA's grading and the AI grading do not match. After investigating these mismatching answers carefully, we found that one of them are obviously the AI grader's error and two of them are the TA's error. The remaining 2 mismatching gradings can be seen okay in both ways using the given rubric. These answers accurately described the process but did not specify the name of the process (DNS query, TCP channel establishment or HTTP request). This disparity occurs because the TA and the AI interpret the meaning of "identify" differently, so both have their reasons. In conclusion, the error rate of the AI in this experiment was only 1/59=1.7%, compared to that of the human TA is 2/59=3.4%

V. CONCLUSION

We have built an AI grader app for college exam using current available LLMs. To validate its effectiveness, we conducted several experiments where LLMs graded mock-up answers from undergraduate network course. We evaluated each LLM's accuracy and precision in statistics from these experiments.

In GPT-3.5 and Gemini-Pro, the grading results varied each time they attempted grading. On the other hands, in GPT-4.0, the grading points generated were consistent over all trials. In the experiment which compared the LLMs' grading with human participants' grading, the accuracy of GPT-4.0's grading surpassed any human grader's. GPT-4.0 understood the rubric better than human graders, and generated the grading points which are more consistent with the mockup answer designer's intention.

With this result of these experiment, we used the AI grader with GPT-4.0 to grade 59 students' answers in sentence. Then this result was compared with the TA's grading. The grading of the TA and the AI matched in 54 students, but did not match in 5 students. As a result of detailed investigation, it was discovered that the AI grader graded only one answers incorrectly, the human TA graded 2 answers incorrectly. Through this result, we concluded that AI-grading with GPT-4.0 is fairly reliable, but not perfect.

VI. DISCUSSION

In the initial state, we tried to create an AI grader app that uses several LLMs for crosschecking the results. As a result of experiments, we concluded that only GPT-4.0 is capable of grading a college exam. In the final experiment, we crosschecked the AI grading with the TA's grading to catch the AI grader's error. However, it may be meaningless in automation if a human grader always grades whenever the AI grader app is used. If an LLM comparable to GPT-4.0 is released, then we can adopt this for crosschecking AI grading.

As we mention above, in the last experiment, the student's typography or grammatical mistake was input intact into the AI grader. The AI grader understood these abnormalities through context and graded it normally. For example, there was an answer where "HTTP" was incorrectly written as "HTT" and another answer where the verb "initiate" was written as "imitate". But the GPT-4.0 understood these as correct words and gave the full points. Such grading of the AI grader is similar to the human TA's grading in the computer science class, which is tolerant to such errors. But if grading requires rigor in spelling and grammar, the rubric should take this aspect into consideration and reflect the grading policy.

For our AI-grading app to generate scores as well as sentences saying the reason for the score, we set the temperature to 0.2 and the maximum token to 300 in OpenAI and Gemini API module. In LLM context, temperature refers a parameter that affects the creativity or randomness of output sentence. Its range is from 0 and 1. The higher the temperature is, the randomness of sentence output increases. It also means that it is more probable that the generated sentence is unreasonable or nonsense. On the other hand, if it is too low, it produces cliché-like sentence output. Therefore, if it is too high, it generate an unreasonable score, and if it is too low, the reason for the grading cannot be properly explained. So, after several initial experiments, we reached this number 0.2. And the maximum token is a parameter that controls the output sentence length. 100 tokens correspond to approximately 75 words [2]. For GPT-4.0 model, both input and output cost tokens, and OpenAI says that they charge $0.03 input per 1000 tokens input, $0.06 output per 1000 tokens [3]. In last experiment, OpenAI charged about $1.00 for grading 59 students' answers.

We used the AI grader in a relatively objective exam, and the result was fairly successful. However, in classes such as humanity or literature, grading using an LLM seems still questionable. The grading in such subjects seems greatly influenced by the individuality or subjectivity of the grader, so it is necessary to explore further whether grading using an LLM will be effective.

VII. References

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